

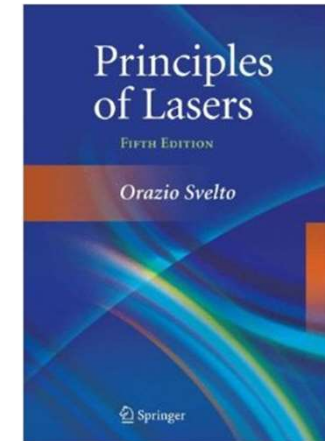


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch2-1 Open Cavity & Gaussian Beam



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Chapter 2: Open Cavity & Gaussian Beam

❖ 2.1 General Questions for Optical Cavity

- Mode & cavity
- Cavity loss
- Classification of optical cavity

❖ 2.2 Matrix Formulation of Geometrical Optics

- ABCD law
- Stable condition for coaxial cavity

2.1 General Question for Optical Cavity

❖ Mode & Cavity

- Optical mode is the photon state that can be distinguished inside the optical cavity
- The resonance mode is fixed once the cavity is given
- Mode theory for optical cavity is the laser mode theory
- Characteristics of mode: energy distribution, frequency, loss, angle of divergence...

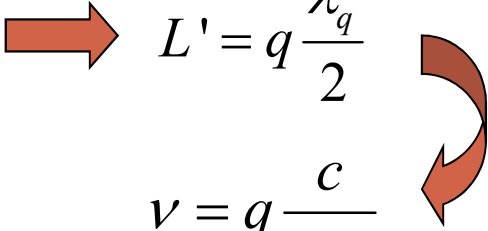
The resonance condition: After one period, it can return to the starting point with the same phase ($= n*2\pi$)

$$\Delta\Phi = \frac{2\pi}{\lambda_q} 2L' = q2\pi \quad \longrightarrow \quad L' = q \frac{\lambda_q}{2}$$

L' : optical length
 λ_q : wavelength in vacuum

$$\nu = q \frac{c}{2L'}$$

$\nu = c/\lambda$



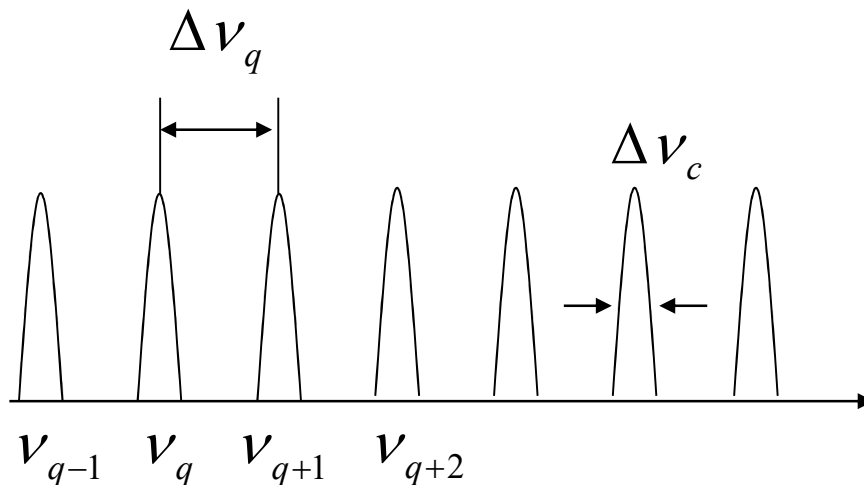
When the cavity is filled with material with refractive index n , then

$$\left. \begin{aligned} L' &= nL \\ \nu_q &= q \frac{c}{2L'} = q \frac{c}{2nL} \end{aligned} \right\} L = q \frac{\lambda'_q}{2}$$

L is the cavity length

λ' is the resonance wavelength in materials

$$\Delta \nu_q = \nu_{q+1} - \nu_q = \frac{c}{2L'}$$



(1) For $L=10$ cm, $n=1$,

$$\Delta \nu_q = 1.5 \times 10^9 \text{ Hz}$$

(2) Smaller cavity length,
the larger longitudinal mode
spacing

Cavity Loss

1. Transmittance Loss (due to low reflection) $\delta = -\frac{1}{2} \ln r_1 r_2$
2. Diffraction Loss (due to diffraction)
3. Geometrical loss (due to angle)
4. Other Loss (material, Q -switch element...)

$$\left. \begin{aligned} \delta &= \delta_1 + \delta_2 + \delta_3 + \delta_4 + \dots \\ I_1 &= I_0 e^{-2\delta} \end{aligned} \right\}$$

$$\delta = \frac{1}{2} \ln \frac{I_0}{I_1} \quad \text{The average loss per cycle}$$

$$\delta' = \frac{1}{2} \frac{I_0 - I_1}{I_0} \quad \text{The percentage loss per cycle}$$

When the cavity loss is very small



$$2\delta' = \frac{I_0 - I_1}{I_0} = \frac{I_0 - I_0 e^{-2\delta}}{I_0} = 1 - \boxed{e^{-2\delta}} \approx 1 - (1 - 2\delta) = 2\delta$$

Parameters related to cavity loss

(1) The average photon lifetime: time needed for the intensity decrease to 1/e

$$I_m = I_0 e^{-2\delta m} \xrightarrow{m = \frac{t}{(2L'/c)}} I_m = I_0 e^{-\frac{2\delta c t}{2L'}} \xrightarrow{\frac{1}{\tau_R} \Rightarrow \boxed{\tau_R = L'/\delta c}} I(t) = I_0 e^{-\frac{t}{\tau_R}}$$

optical length

density of photon number $\xrightarrow{I \propto N}$

$$N(t) = N_0 e^{-\frac{t}{\tau_R}}$$

$$-dN = \frac{N_0}{\tau_R} e^{-\frac{t}{\tau_R}} dt$$

$$\bar{t} = \frac{1}{N_0} \int (-dN) t = \frac{1}{N_0} \int_0^\infty t \left(\frac{N_0}{\tau_R} \right) e^{-\frac{t}{\tau_R}} dt = \left(\frac{1}{\tau_R} \right) \int_0^\infty t e^{-\frac{t}{\tau_R}} dt = \tau_R$$

$$\boxed{\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}}$$

Parameters related to cavity loss

(2) Q -factor for passive cavity

$$Q = 2\pi\nu \frac{\text{Total energy in the cavity } (\xi)}{\text{Energy loss per unit time } (P)}$$

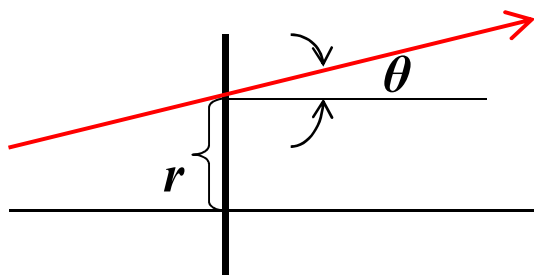
$$\begin{array}{l} \xi = Nh\nu V \\ P = -h\nu V \frac{dN}{dt} \end{array} \quad \xrightarrow{N = N_0 e^{-\frac{t}{\tau_R}}} \quad Q = 2\pi\nu \frac{L'}{\delta c} = \omega\tau_R$$

$$Q \propto \tau_R \propto \frac{1}{\delta}$$

The smaller the cavity loss, the higher the Q -factor of the cavity

2.2 Matrix Formulation of Geometrical Optics

1. Formulation of Geometrical Optics (ABCD matrix)

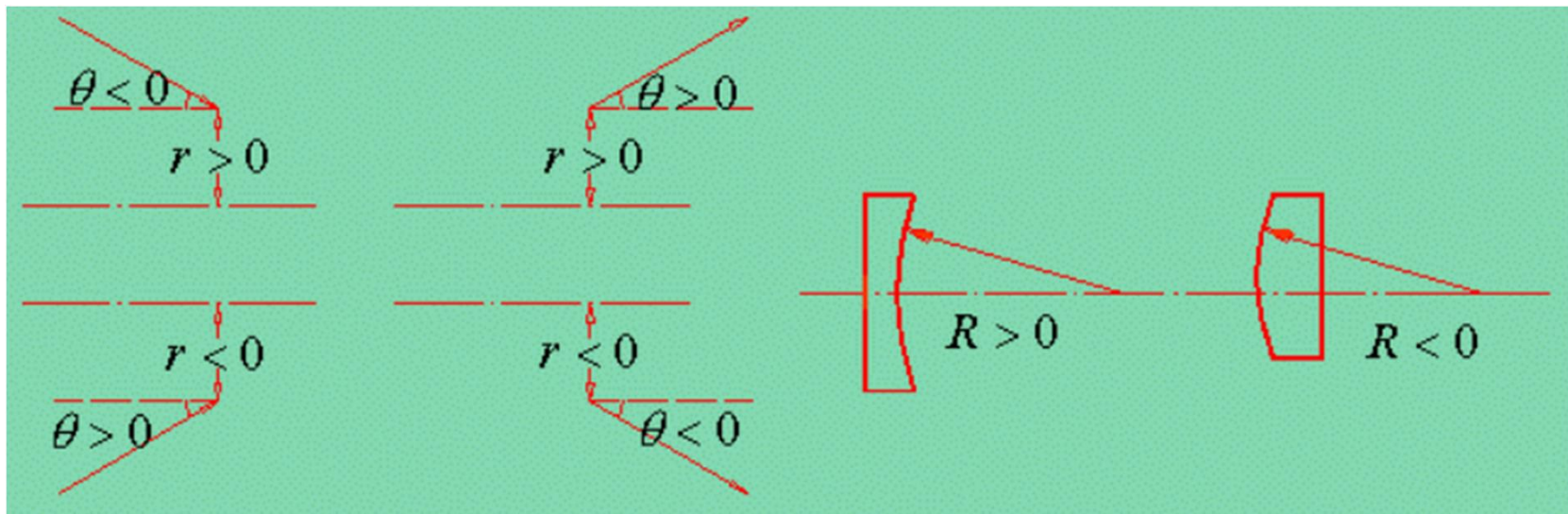


A ray can be described by

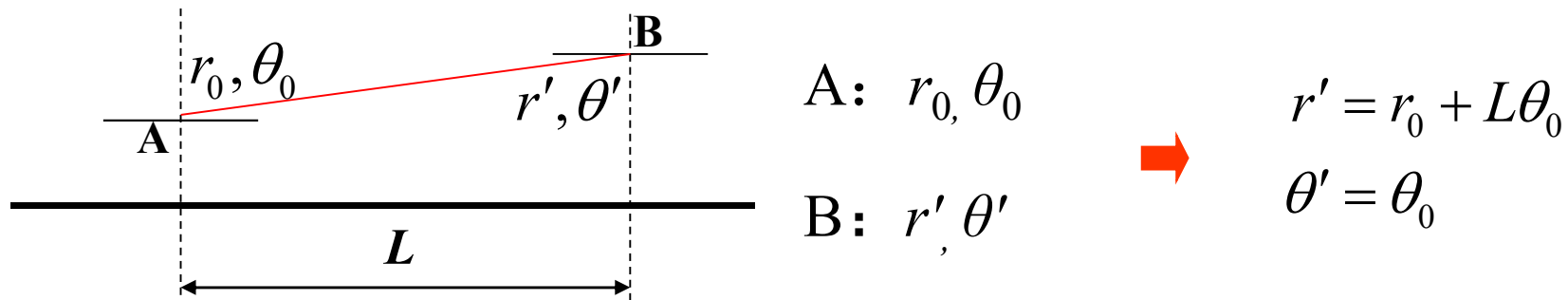
r — distance with optical axis

θ — angle with optical axis

Small angle (paraxial-ray) $\text{tg} \theta \approx \sin \theta \approx \theta$



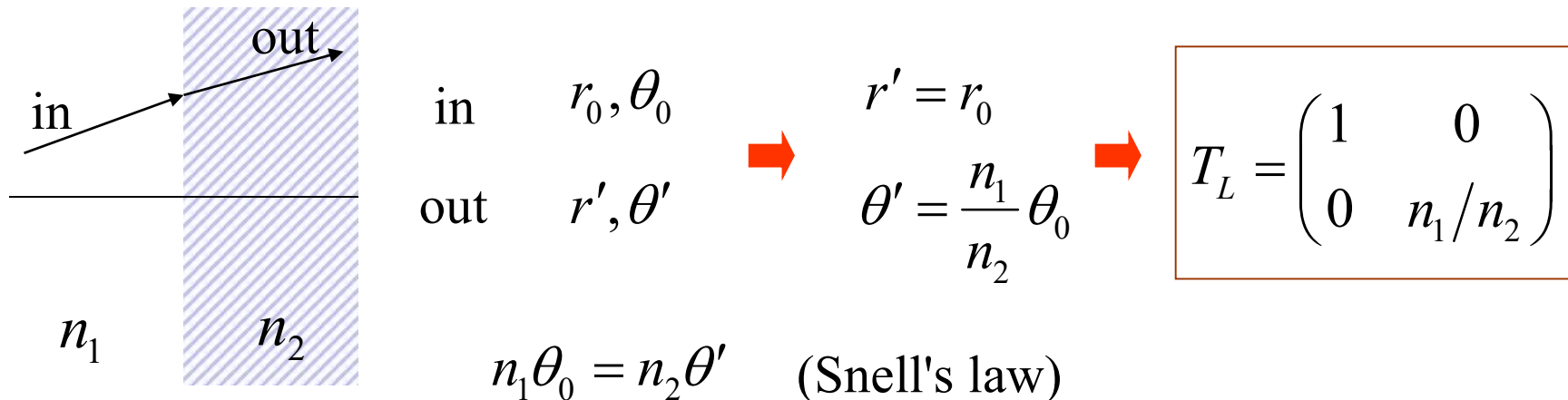
2. The matrix in free space



Free Space Ray Matrix

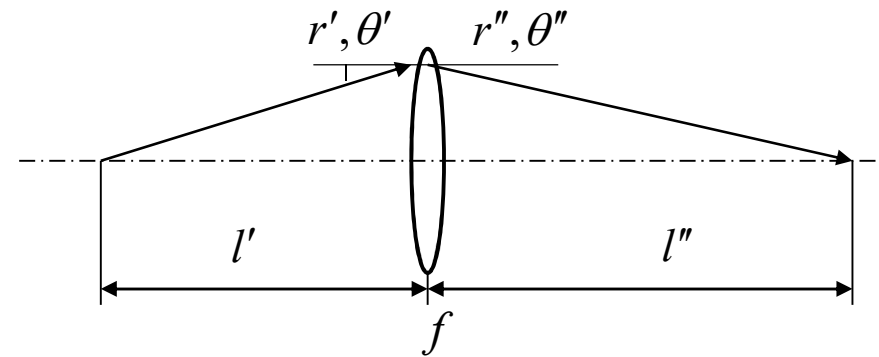
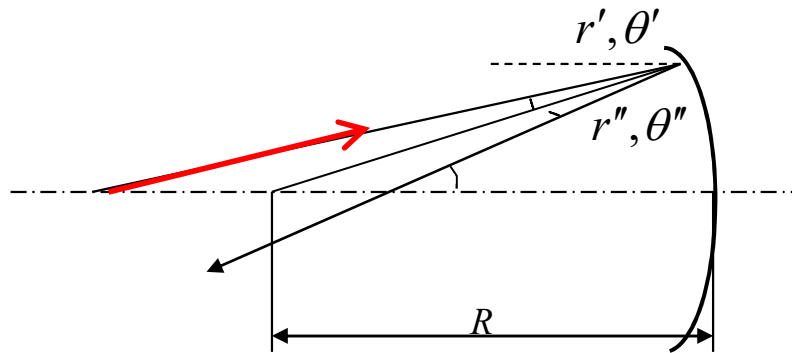
$$\begin{pmatrix} r' \\ \theta' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = T_L \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

3. Interface between air and dielectric (refractive index n_2)



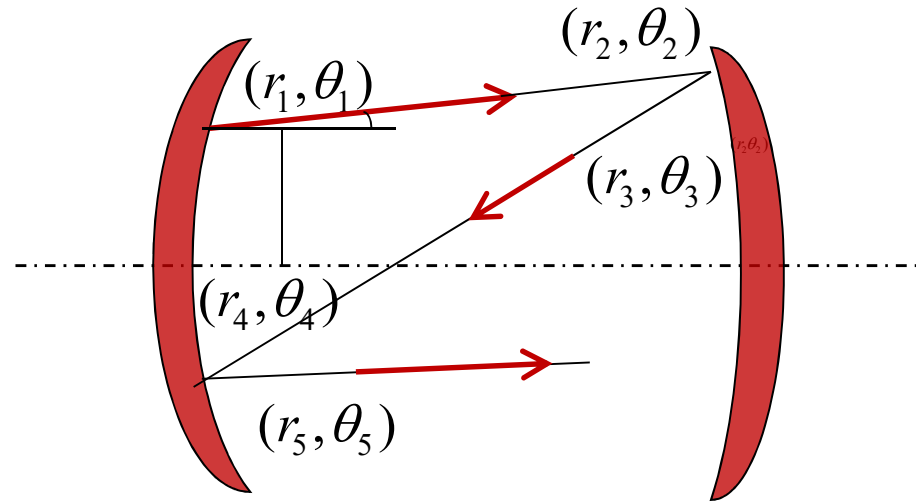
4. The matrix for spherical mirror (and lens)

$$\begin{aligned}
 r'' &= r' \\
 \theta'' &= \theta' - \frac{2}{R} r' \quad \Rightarrow \quad \begin{pmatrix} r'' \\ \theta'' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r' \\ \theta' \end{pmatrix} \Rightarrow T_R = \begin{pmatrix} 1 & 0 \\ -2/R & 1 \end{pmatrix} \quad \text{Planer mirror} \\
 & \quad \quad \quad T_R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
 \end{aligned}$$



$$\begin{aligned}
 r'' &= r' & \theta' &= r'/l' \\
 -\theta'' &= r''/l'' & & \\
 \theta' - \theta'' &= r' \left(\frac{1}{l'} + \frac{1}{l''} \right) = r' \frac{1}{f} & \Rightarrow & \quad r'' = r' \\
 & & \Rightarrow & \quad \theta'' = -\frac{1}{f} r' + \theta' \\
 & & & \Rightarrow T_f = \begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix} \quad f = \frac{R}{2}
 \end{aligned}$$

The two are the same



The first travel
$$\begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} = T_L \begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

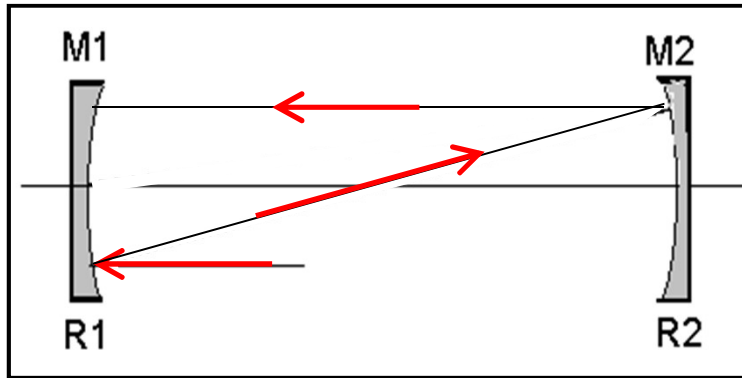
The second travel
$$\begin{pmatrix} r_3 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_2} & 1 \end{pmatrix} \begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} = T_L \begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_2} & 1 \end{pmatrix}$$

The third travel
$$\begin{pmatrix} r_4 \\ \theta_4 \end{pmatrix} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_3 \\ \theta_3 \end{pmatrix} = T_L \begin{pmatrix} r_3 \\ \theta_3 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

The fourth travel
$$\begin{pmatrix} r_5 \\ \theta_5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_1} & 1 \end{pmatrix} \begin{pmatrix} r_4 \\ \theta_4 \end{pmatrix} = T_L \begin{pmatrix} r_4 \\ \theta_4 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_1} & 1 \end{pmatrix}$$

5. ABCD Matrix Application (resonance inside spherical mirror)

球面镜腔中往返一周的光线矩阵（简称往返矩阵）



$$\begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} = T_{R_1} T_L T_{R_2} T_L \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} \\ = T \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2/R_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2/R_2 & 1 \end{pmatrix} \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

$$A = 1 - \frac{2L}{R_2}$$

$$C = - \left[\frac{2}{R_1} + \frac{2}{R_2} \left(1 - \frac{2L}{R_1} \right) \right]$$

$$B = 2L \left(1 - \frac{L}{R_2} \right)$$

$$D = - \left[\frac{2L}{R_1} - \left(1 - \frac{2L}{R_1} \right) \left(1 - \frac{2L}{R_2} \right) \right]$$

If travel n rounds

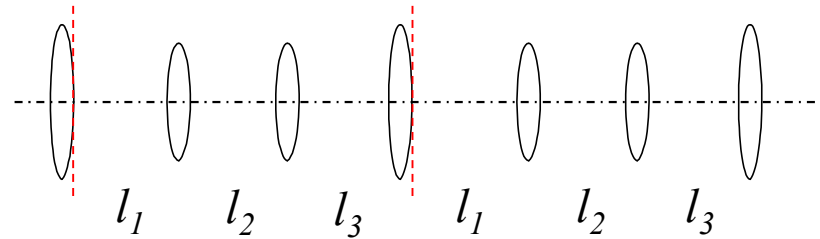
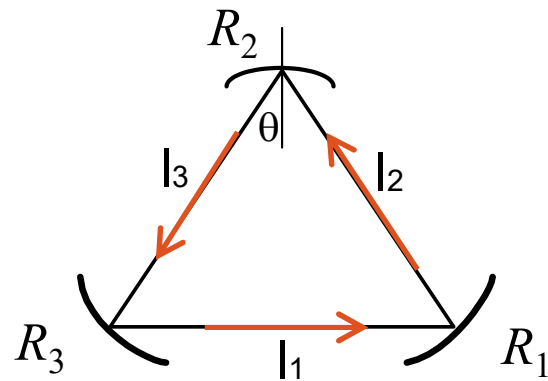
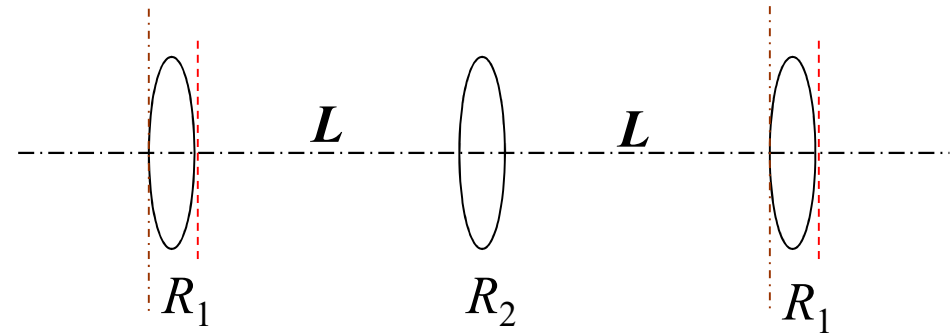
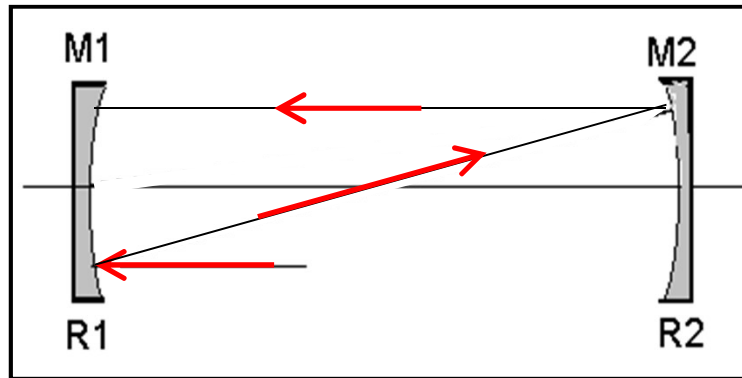
$$\begin{pmatrix} r_n \\ \theta_n \end{pmatrix} = \underbrace{T \dots T}_n \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = T_n \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

$$T_n = \frac{1}{\sin \varphi} \begin{pmatrix} A \sin n\varphi - \sin(n-1)\varphi & B \sin n\varphi \\ C \sin n\varphi & D \sin n\varphi - \sin(n-1)\varphi \end{pmatrix}$$

where $\varphi = \arccos \frac{1}{2}(A + D)$

- Light travel in cavity can be treated as light pass through series of optical components
- The matrix is not related to their starting position and light travel direction
- **General Steps** (1) find the optical loop (2) fix the period (3) write down each component

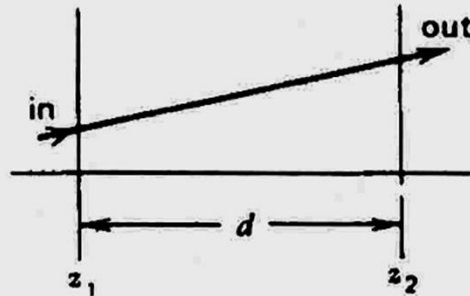
Spherical cavity = series of lens



$$\begin{pmatrix} 1 & 0 \\ -2/R_3 & 1 \end{pmatrix} \begin{pmatrix} 1 & l_3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2/R_2 & 1 \end{pmatrix} \begin{pmatrix} 1 & l_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2/R_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & l_1 \\ 0 & 1 \end{pmatrix}$$

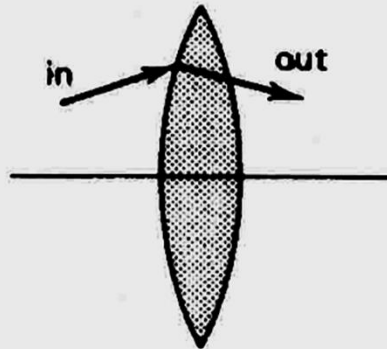
Ray matrices for some common cases

- (1) Homogeneous Medium:
Length d



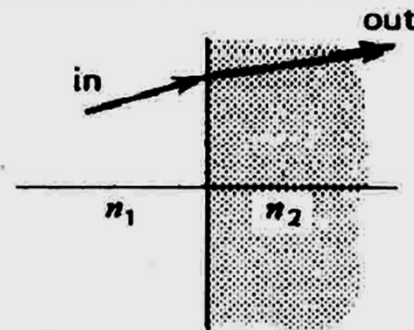
$$\begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$$

- (2) Thin Lens:
Focal length f
($f > 0$, converging;
 $f < 0$, diverging)



$$\begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$$

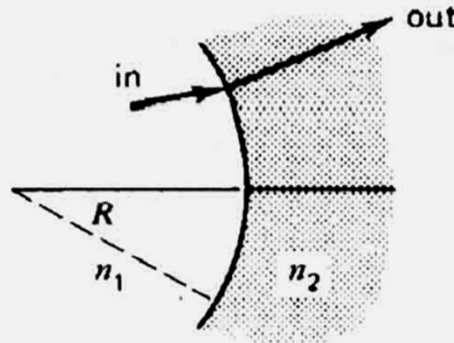
- (3) Dielectric Interface:
Refractive indices
 n_1, n_2



$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix}$$

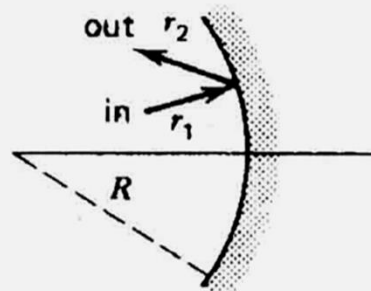
Ray matrices for some common cases

- (4) Spherical Dielectric Interface:
Radius R



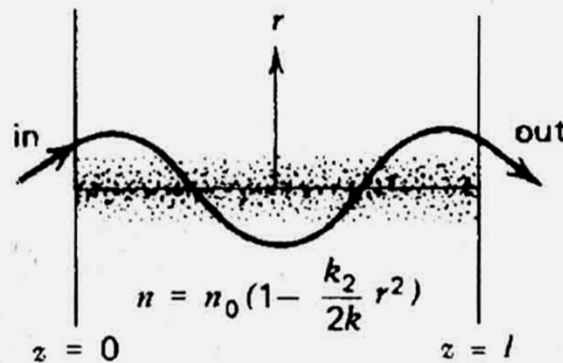
$$\begin{bmatrix} 1 & 0 \\ \frac{n_2 - n_1}{n_2} \frac{1}{R} & \frac{n_1}{n_2} \end{bmatrix}$$

- (5) Spherical Mirror:
Radius of curvature R



$$\begin{bmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{bmatrix}$$

- (6) A medium with a quadratic index profile

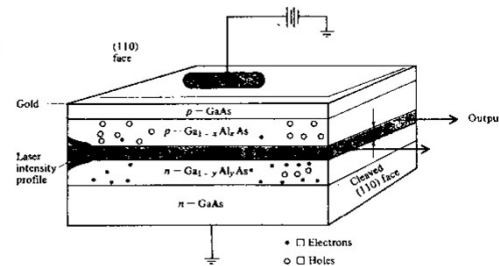
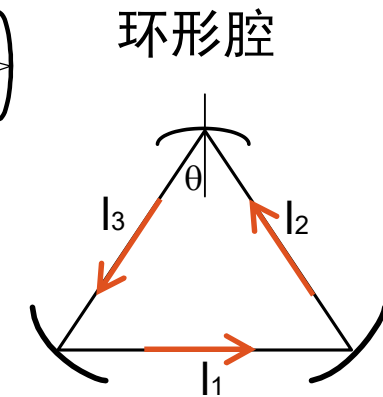
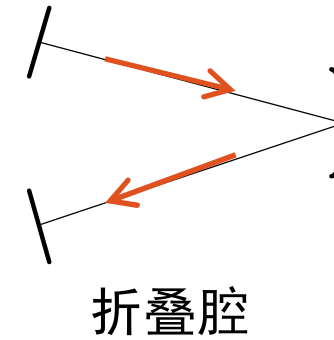
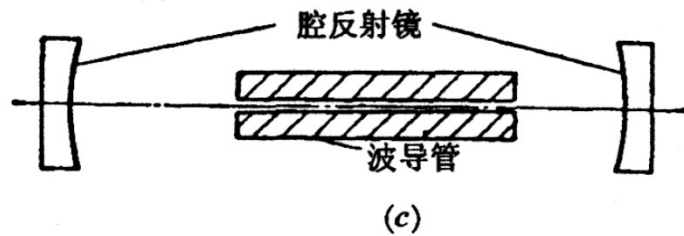
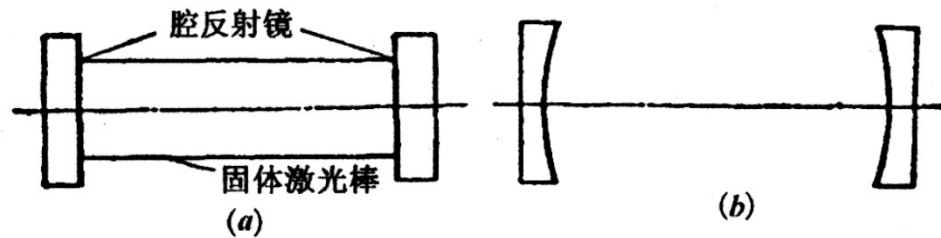


$$\begin{bmatrix} \cos\left(\sqrt{\frac{k_2}{k}} l\right) & \sqrt{\frac{k}{k_2}} \sin\left(\sqrt{\frac{k_2}{k}} l\right) \\ -\sqrt{\frac{k_2}{k}} \sin\left(\sqrt{\frac{k_2}{k}} l\right) & \cos\left(\sqrt{\frac{k_2}{k}} l\right) \end{bmatrix}$$

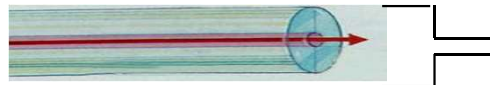
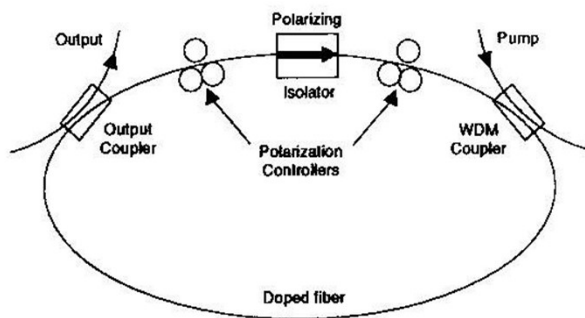
平方折射率介质

Classification of Optical Cavity

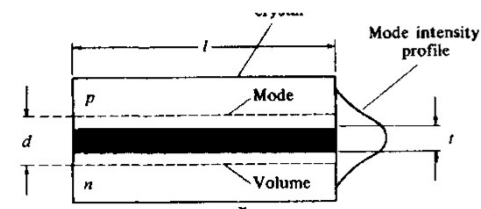
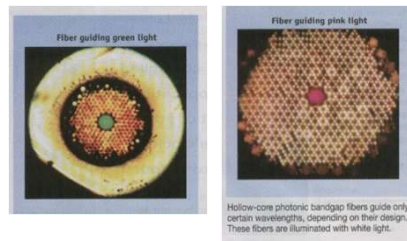
1. Construction



dielectric waveguide

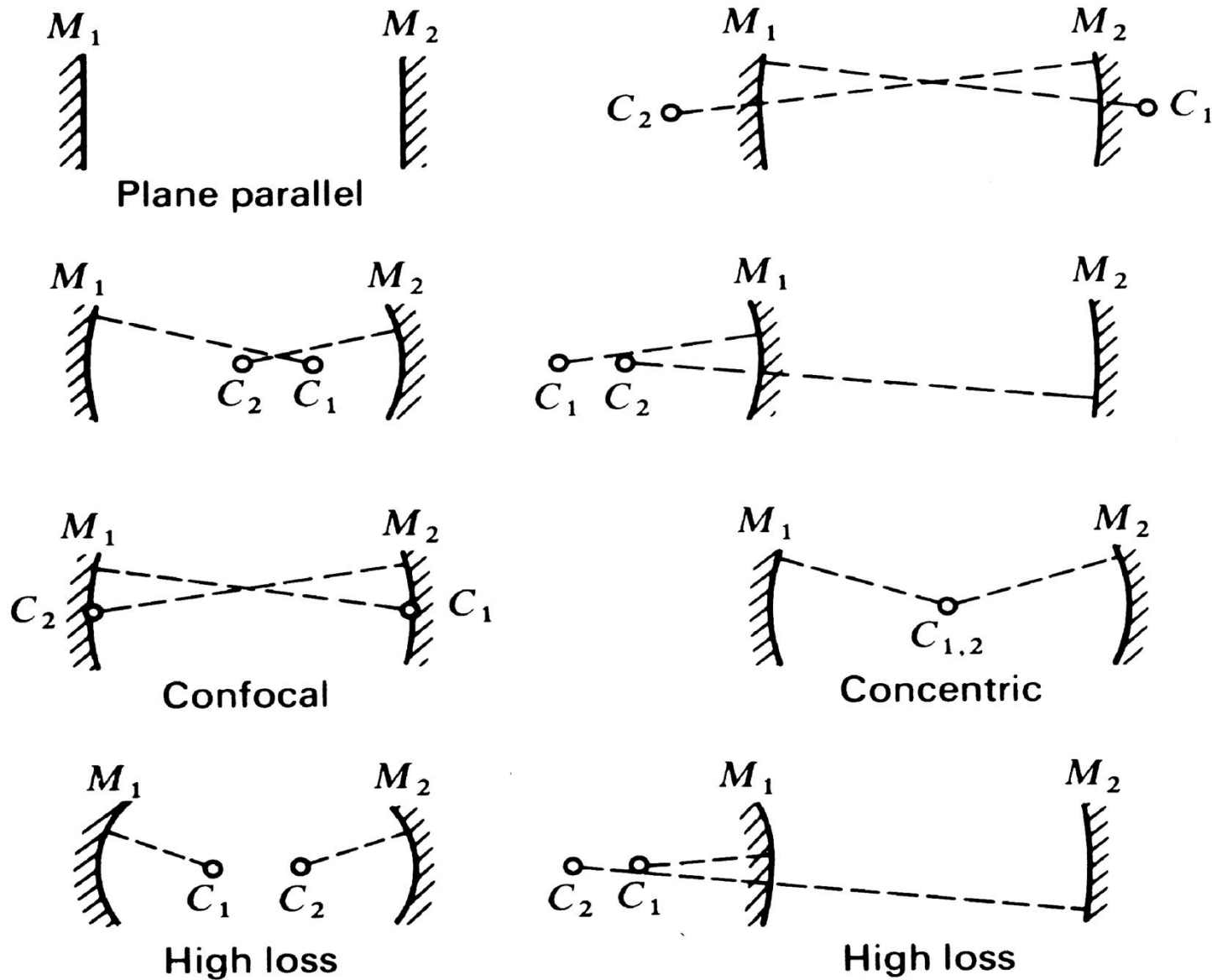


semiconductor laser



$$\begin{aligned} \eta_2 &> \eta_1, \\ \eta_2 &> \eta_3 \end{aligned}$$

2. Geometrical leakage loss: stable, semi-stable, critical



3. Stable condition for spherical mirror cavity

- Co-axis spherical mirror cavity: two mirrors are spherical mirror, have the same optical axis.

concave mirror $R > 0$; convex mirror $R < 0$; plane mirror $R = \infty$
凹透镜 凸透镜 平面镜

- Stable Condition: Geometrical loss

Stable Cavity: Any light with small angle can be reflected all the times inside the cavity and can not escape from it $\Leftrightarrow$ small geometrical loss (low loss cavity)

Unstable Cavity: Light with small angle will escape out of the cavity after certain periods $\Leftrightarrow$ large geometrical loss (high loss cavity)

- Ray matrix can be used to analyze the geometrical loss and determine whether the cavity is stable

If travel n rounds

$$\begin{pmatrix} r_n \\ \theta_n \end{pmatrix} = \underbrace{T \dots T}_n \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = T_n \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

$$T_n = \frac{1}{\sin \varphi} \begin{pmatrix} A \sin n\varphi - \sin(n-1)\varphi & B \sin n\varphi \\ C \sin n\varphi & D \sin n\varphi - \sin(n-1)\varphi \end{pmatrix}$$

where $\varphi = \arccos \frac{1}{2}(A + D)$

For stale cavity



$$\frac{1}{2}|A + D| < 1$$

$A_n B_n C_n D_n$ is infinite, then light can not escape from the cavity after reflected for n times. φ is a real number, and not equal to $k\pi$

Discussion $\varphi = \arccos \frac{1}{2}(A+D)$

$$(1) \quad \left| \frac{1}{2}(A+D) \right| < 1 \Rightarrow \varphi \text{ Real } A_n, B_n, C_n, D_n \text{ is limited} \Rightarrow \text{Stable}$$

$$(2) \quad \left| \frac{1}{2}(A+D) \right| > 1 \Rightarrow \varphi \text{ Image } n \uparrow \rightarrow A_n, B_n, C_n, D_n \uparrow \Rightarrow \text{Unstable}$$

$$(3) \quad \left| \frac{1}{2}(A+D) \right| = 1 \quad \Leftrightarrow \frac{1}{2}(A+D) = \pm 1 \Rightarrow \varphi = k\pi \quad \begin{cases} k \text{ odd } (-1) \\ k \text{ even } (+1) \end{cases}$$

Find the limitation

$$k \text{ odd} \quad \frac{1}{2}(A+D) = -1 \quad T_n = (-1)^{n+1} \begin{pmatrix} An + (n-1) & Bn \\ Cn & Dn + (n-1) \end{pmatrix}$$

$$k \text{ even} \quad \frac{1}{2}(A+D) = +1 \quad T_n = 1^{(n+1)} \begin{pmatrix} An - (n-1) & Bn \\ Cn & Dn - (n-1) \end{pmatrix}$$

Another Expression

$$\because A = 1 - \frac{2L}{R_2} \quad D = -\left[\frac{2L}{R_1} - \left(1 - \frac{2L}{R_1}\right) \left(1 - \frac{2L}{R_2}\right) \right]$$

$$\frac{1}{2}(A + D) \equiv 1 - \frac{2L}{R_1} - \frac{2L}{R_2} + \frac{2L^2}{R_1 R_2}$$

$$-1 < \frac{1}{2}(A + D) < 1 \quad \Rightarrow \quad 0 < \left(1 - \frac{L}{R_1}\right) \left(1 - \frac{L}{R_2}\right) < 1 \quad \Rightarrow \quad 0 < g_1 g_2 < 1$$

$$g_1 = 1 - \frac{L}{R_1}$$
$$g_2 = 1 - \frac{L}{R_2}$$

Stable Cavity

$$0 < g_1 g_2 < 1$$

Unstable Cavity

$$g_1 g_2 > 1, \quad g_1 g_2 < 0$$

Critical Cavity

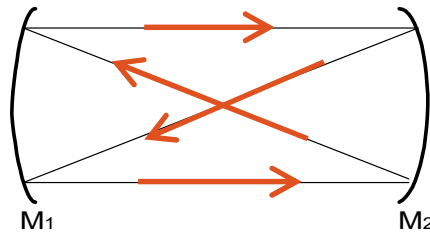
$$g_1 g_2 = 0, \quad g_1 g_2 = 1$$

Some Examples

1. Planer mirror cavity ($R = \infty$)

$$T_n = \begin{pmatrix} 1 & 2Ln \\ 0 & 1 \end{pmatrix} \rightarrow \begin{cases} r_n = r_0 + 2Ln\theta_0 \\ \theta_n = \theta_0 \end{cases} \rightarrow \begin{cases} \theta_0 \neq 0 & r_n \uparrow & \text{Unstable} \\ \theta_0 = 0 & r_n = r_0 & \text{Stable} \end{cases}$$

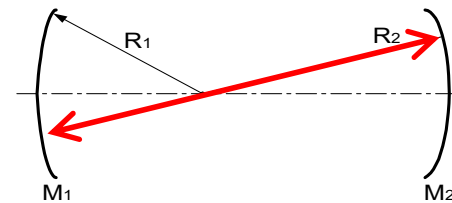
2. Symmetrical confocal cavity $R_1=R_2=L$



$$T_n = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Form close loop after two period: **Stable**

3. Concentric cavity $R_1+R_2=L$



Critical

$$\left| \frac{1}{2}(A+D) \right| < 1 \text{ Stable; } \left| \frac{1}{2}(A+D) \right| > 1 \text{ Unstable; } \left| \frac{1}{2}(A+D) \right| = 1 \text{ Criticle}$$

It is used for all kinds of cavity. You can determine the stability by the ray matrix.